

Conducting Multiple Linear Regression in R

A Practical Handout for Data Analysis

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Regression analysis is widely used in statistics James et al. (2021).

The analysis was performed using R R Core Team (2024).

Econometric models are discussed by Wooldridge Wooldridge (2019).

1. Introduction

Multiple linear regression (MLR) is one of the most widely used statistical techniques in quantitative research. It models the relationship between a continuous outcome variable and two or more predictor variables. Unlike simple linear regression with a single predictor, MLR allows researchers to control multiple factors simultaneously, giving a more nuanced understanding of which variables explain variation in the outcome.

The general form of a multiple linear regression model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$$

where Y is the dependent variable, X_1 through X_k are independent (predictor) variables, β_0 is the intercept, β_1 through β_k are the regression coefficients, and ϵ is the error term capturing unexplained variation.

This handout complements the presentation by providing additional theoretical context, extended R code examples, and references for further study. It assumes familiarity with basic regression theory and focuses on the practical implementation in R.

2. Data Preparation in R

In R Before running a regression, data must be loaded into R and inspected for issues such as missing values, incorrect variable types, or outliers. R provides several built-in functions for this purpose.

2.1 Creating Data

Or Loading Data For demonstration purposes, data can be created directly in R using `data.frame()`. In practice, data is often imported from CSV files using `read.csv()` or from Excel using `read_xl::read_excel()`. The example below creates a small dataset with salary as the outcome and age, education, and experience as predictors:

```
data <- data.frame(
  Salary      = c(2000, 2500, 3000, 3500, 4000),
  Age         = c(22, 25, 30, 35, 40),
  Education   = c(12, 14, 16, 16, 18),
  Experience  = c(1, 3, 5, 7, 10)
)
```

Example

A researcher studying income determinants could collect survey data with annual salary, years of schooling, age, and work experience from 500 employees, then import it with

```
read.csv("salary_data.csv")
```

2.2 Inspecting the Data

After loading data, the following functions help detect potential issues before modeling:

```
head(data) # first 6 rows — confirms data loaded correctly
```

```
summary(data) # min, max, mean, quartiles per variable
```

```
str(data) # variable types: numeric, factor, character, etc.
```

```
is.na(data) # check for missing values
```

Example

Running `summary(data)` on the salary dataset would reveal that average salary is 3000, education ranges from 12 to 18 years, and experience from 1 to 10 years — useful context before modeling.

3. Model Specification and Estimation

In R, regression models are specified using a formula interface. The tilde (`~`) separates the dependent variable (left) from the predictors (right). The `lm()` function fits the model using ordinary least squares (OLS).

Model formula: $\text{Salary} \sim \text{Age} + \text{Education} + \text{Experience}$

```
model <- lm(Salary ~ Age + Education + Experience, data = data)
```

The `lm()` call returns a model object stored in `model`. This object contains all estimated coefficients, residuals, fitted values, and diagnostic information. To interact with it, several extractor functions are available:

```
fitted(model)

residuals(model)

summary(model)
```

Example

If a company wants to predict what salary to offer to a 33-year-old with 16 years of education and 6 years of experience, they specify the model above, then use `predict()` with a new data frame.

4. Interpreting the Output

The `summary(model)` output provides the core regression results. The key components and their interpretation are:

Component Interpretation Estimate (coefficient) Expected change in Y for a one-unit increase in X, holding all other predictors constant. E.g., if $\beta = 50$ for Age, each additional year of age is associated with €50 more salary. Std. Error Measures the precision of the coefficient estimate. Smaller values indicate more reliable estimates. t value Ratio of the estimate to its standard error. Larger absolute values indicate stronger evidence against the null hypothesis $\beta = 0$. $\Pr(>|t|)$ p-value for the two-sided test $H_0: \beta = 0$. Typically, $p < 0.05$ is considered statistically significant. R-squared Proportion of variance in Y explained by the model. Ranges from 0 to 1; higher values indicate better fit. Adjusted R² R² penalized for the number of predictors. Preferable to R² when comparing models with different numbers of predictors. F-statistic Tests whether at least one predictor has a non-zero coefficient (overall model significance).

Example

In the salary model, a coefficient of 100 on Experience means that for each additional year of experience—holding age and education constant—the predicted salary increases by €100. If the p-value for Education is 0.03, we conclude education has a statistically significant effect at the 5% level.

5. Model Diagnostics

After estimation, it is essential to check whether the OLS assumptions are reasonably met. R's `plot(model)` generates four standard diagnostic plots:

Residuals vs. Fitted:

Checks linearity. Ideally, residuals scatter randomly around zero with no pattern. A curved pattern suggests a non-linear relationship. • Normal Q-Q Plot: Checks normality of residuals. Points should fall approximately on the diagonal line. Heavy deviations indicate non-normal errors. • Scale-Location (Spread-Location): Checks homoscedasticity (constant variance). A horizontal band with randomly scattered points is ideal. A funnel shape indicates heteroscedasticity. • Cook's Distance / Residuals vs. Leverage: Identifies influential observations. Points with high leverage and large residuals can disproportionately affect the regression line.

```
par(mfrow = c(2, 2))
```

```
# display all 4 plots in a 2x2 grid plot(model)
```

Example

When modeling exam scores from study hours and sleep, a Scale-Location plot showing a widening band at higher fitted values would suggest that students with predicted high scores also show more variable results — violating homoscedasticity and potentially requiring a log transformation or weighted regression.

6. Making Predictions

Once a model is fitted, it can be used to generate predictions for new observations using the `predict()` function. A new data frame with the same predictor names must be supplied:

```
new_employee <- data.frame(  
  Age      = 33,  
  Education = 16,  
  Experience = 6  
)
```

```
predict(model, newdata = new_employee)
```

With 95% prediction interval:

```
predict(model, newdata = new_employee, interval = "prediction", level = 0.95)
```

Example: An HR manager who has fitted the salary regression can enter the profile of a new job applicant (age 33, 16 years education, 6 years' experience) and obtain a predicted salary alongside a 95% prediction interval, giving both a point estimate and a range for decision-making.

7. Class Exercise

Complete the following exercise using R. The goal is to practice specifying and interpreting a multiple linear regression model with your own variables.

Task:

- Create a data frame with at least 8 observations and at least two predictor variables (e.g., house price predicted by size and number of rooms, or student GPA predicted by study hours and attendance rate).
- Fit a multiple linear regression model using #

```
lm(y ~ x1 + x2, data = your_data)
```

- Use `summary()` to examine the output and answer the following questions:
- Which predictors are statistically significant at the 5% level?
- What is the adjusted R-squared, and what does it tell you about model fit?
- Interpret at least one coefficient in plain language.
- Run `plot(model)` and describe what each diagnostic plot indicates about the validity of regression assumptions.
- Use `predict()` to estimate the outcome for a new observation of your choice.

Starter template:

1. Create your own dataset

```
my_data <- data.frame(  
  y = c(...), # your outcome variable  
  x1 = c(...), # first predictor  
  x2 = c(...) # second predictor  
)
```

2. Inspect

```
head(my_data)

summary(my_data)
```

3. Fit model

```
my_model <- lm(y ~ x1 + x2, data = my_data)
```

4. Results

```
summary(my_model)
```

5. Diagnostics

```
par(mfrow = c(2, 2))

plot(my_model)
```

6. Predict for a new observation

```
new_obs <- data.frame(
  x1 = ...,
  x2 = ...
)

predict(my_model, newdata = new_obs)
```

8. Comprehensive End-to-End Example

The following example walks through the entire workflow from data creation to prediction in a single, self-contained R script. This mirrors what was covered across all slides of the presentation and can be used as a reference template.

Scenario

A company collects data on 5 employees and wants to model monthly salary (in €) as a function of age, years of education, and years of work experience.

STEP 1: Load / create data

```
data <- data.frame(  
  Salary      = c(2000, 2500, 3000, 3500, 4000),  
  Age         = c(22, 25, 30, 35, 40),  
  Education   = c(12, 14, 16, 16, 18),  
  Experience  = c(1, 3, 5, 7, 10)  
)
```

STEP 2: Inspect

```
head(data)  
summary(data)  
str(data)
```

STEP 3: Specify and estimate model

```
model <- lm(Salary ~ Age + Education + Experience, data = data)
```

STEP 4: Extract results

```
summary(model)
```

Key output: coefficients, p-values, R-squared, F-statistic

STEP 5: Run diagnostics

```
par(mfrow = c(2, 2))  
plot(model)
```

Check: linearity, normality, homoscedasticity, influential points

STEP 6: Predict for a new employee

```
new_employee <- data.frame(  
  Age      = 33,  
  Education = 16,  
  Experience = 6  
)
```

```
predict(model, newdata = new_employee) predict(model, newdata = new_employee, interval  
= "prediction", level = 0.95)
```

Result:

point estimate + 95% prediction interval for salary

This six-step workflow load, inspect, specify, estimate, diagnose, predict constitutes the core practical competency in applied regression analysis using R. Mastery of this sequence enables researchers to efficiently analyze relationships between variables and communicate results with statistical rigor.

Affidavit

I hereby affirm that this submitted paper was authored unaided and solely by me. Additionally, no other sources than those in the reference list were used. Parts of this paper, including tables and figures, that have been taken either verbatim or analogously from other works have in each case been properly cited with regard to their origin and authorship.

This paper either in parts or in its entirety, be it in the same or similar form, has not been submitted to any other examination board and has not been published.

I acknowledge that the university may use plagiarism detection software to check my thesis. I agree to cooperate with any investigation of suspected plagiarism and to provide any additional information or evidence requested by the university.

Checklist:

- The handout contains 3-5 pages of text.
- The submission contains the Quarto file of the handout.
- The submission contains the Quarto file of the presentation.
- The submission contains the HTML file of the handout.
- The submission contains the HTML file of the presentation.
- The submission contains the PDF file of the handout.
- The submission contains the PDF file of the presentation.
- The title page of the presentation and the handout contain personal details (name, email, matriculation number).
- The handout contains a bibliography, created using BibTeX with an APA citation style.
- Either the handout or the presentation contains R code that proves the expertise in coding.
- The filled out Affidavit.
- The link to the presentation and the handout published on GitHub.

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Date: 26 May 2026

Place: Cologne

References

- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning* (2nd ed.). Springer. <https://doi.org/10.1007/978-1-0716-1418-1>
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